Lecture 1 v2:
Introduction and Review of Probability

25117 - Econometrics

Universitat Pompeu Fabra

October 14, 2026

About this course

Evaluation:

- *Participation:* 5% + *Midterm 1:* 15% + *Midterm 2:* 20% + *Cumulative Final Exam:* 60%

What's Econometrics about?

- Economics is the social science that studies how economic agents (i.e., individuals, businesses, governments, societies, etc.) allocate scarce resources (e.g., time, money, etc.) to satisfy an objective.

- Resulting allocation choices matter — *"There is no free lunch!"* — as they suggest important relationships (e.g., "Education matters for future income/personal well-being/etc.")

- Econometrics is a specialized branch of economics that focuses on the application of statistical methods to analyze and quantify these economic relationships.

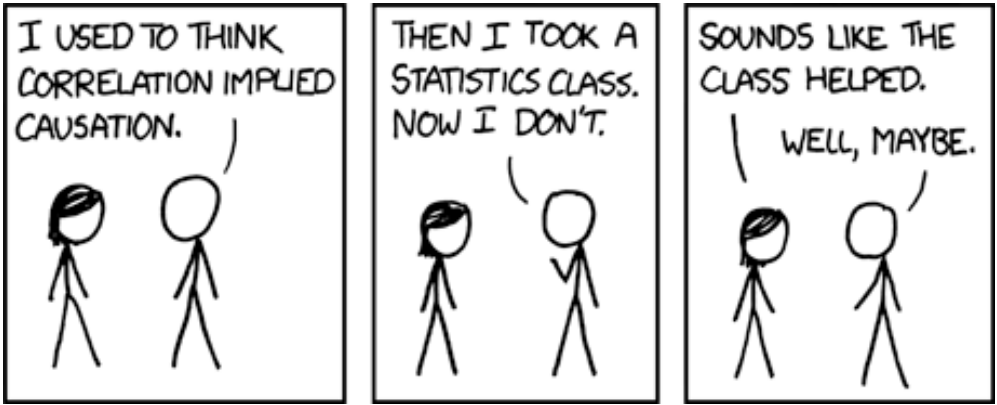
  → How does another year of education change earnings?
  → What is the quantitative effect of reducing class size on student achievement?
  → What is the effect on housing prices of environmental improvements?

- Goal: To bridge the gap between economic theory and real-world data by using empirical evidence to test and refine economic theories.

Inferring Causal Effects (and more)

- In most applied econometrics problems, we seek to identify the causal effect of a change in a treatment variable D on an outcome Y.
- Ideally, this would require a randomized control experiment: the treatment D is randomly assigned to a sample population.
- Randomization ensures that there are no systematic differences between the control (or comparison) and the treated groups.
- Most of these experiments would be completely unethical (and if not, extremely costly) !
- Instead, we have to rely on observational (i.e., non-experimental) data, which might be jointly determined. How to identify causal effects, then?

A sentence you are going to hear a lot



Course Outline

BLOCK 1

1. Introduction and Review of Probability
2. Review of Statistics
3. Simple Linear Regression
4. OLS Implementation

⟶ Midterm 1

BLOCK 2

5. Multivariate Linear Regression
6. Inference and Interpretation
7. Functional Forms

⟶ Midterm 2

BLOCK 3

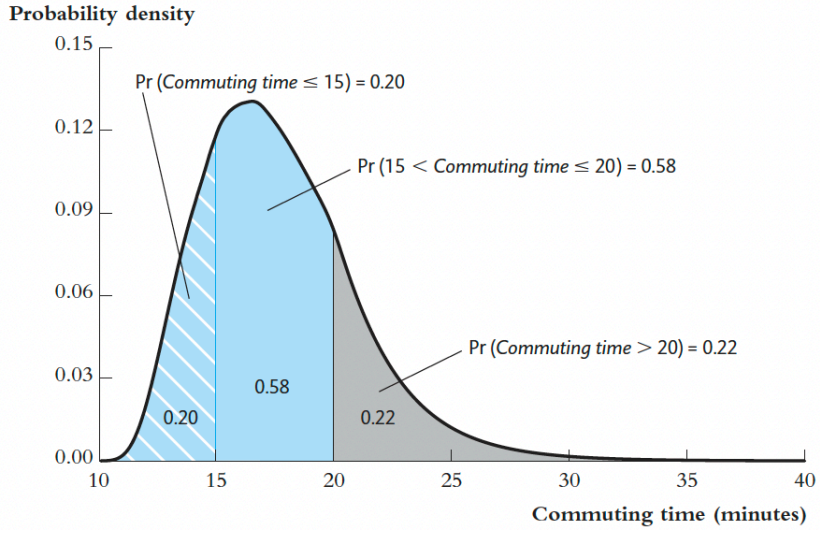
8. Threats to Identification
9. Instrumental Variables
10. Panel Data
11. Binary Choice Models

⟶ Cumulative Final Exam

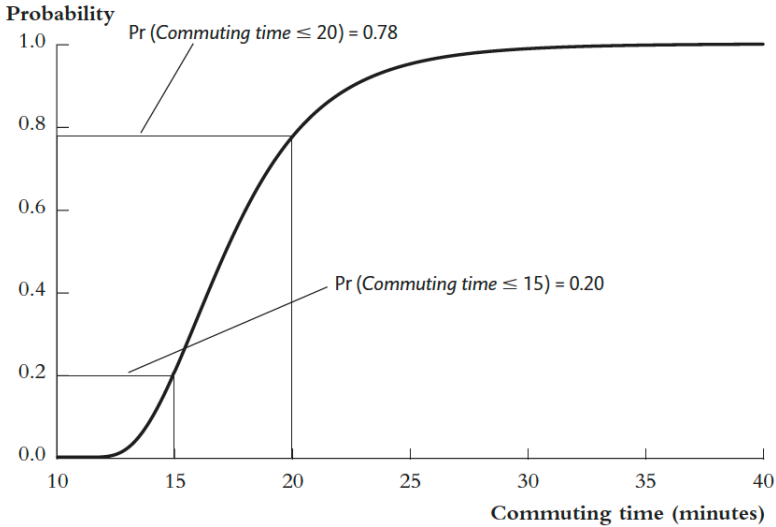
Some Revision

1. A random variable is a mathematical concept used to describe and model uncertain events
   - NB: A random variable isn't inherently "unpredictable"; it represents different values tied to uncertain outcomes, each with assigned probabilities
2. The mutually exclusive potential results of a random process are called the outcomes
3. The set of all possible outcomes is called the sample space. A subset of the sample space is called an event.
4. The probability distribution of a random variable, X, is the list of all possible values, x, of the variable and the probability that each value will occur, $\Pr(X = x)$. These probabilities sum to 1.
   - NB: We talk about probability density function for continuous random variables (p.d.f.)
5. The cumulative distribution function (c.d.f.) is the probability that the random variable is less than or equal to a particular value

P.d.f. of a continuous random variable



C.d.f. of a continuous random variable



Statistical Moments

1. The mean (or expected value) indicates the long-run average of a random variable Y:

$$\mathbb{E}(Y) \equiv \mu_Y = \sum_{i=1}^{k} y_i \Pr(Y = y_i)$$

2. The standard deviation (resp., variance) informs about the (resp., squared) dispersion of Y:

$$\sigma_Y^2 \equiv \mathrm{Var}(Y) = \mathbb{E}[(Y - \mu_Y)^2] = \sum_{i=1}^{k} (y_i - \mu_Y)^2 \Pr(Y = y_i)$$

3. The skewness informs about the asymmetry of Y:

$$\gamma_Y = \mathbb{E}[(Y - \mu_Y)^3] \sigma_Y^{-3}$$

4. The kurtosis of Y is a measure of how much mass is in its tails (i.e., chance of extreme value):

$$\kappa_Y = \mathbb{E}[(Y - \mu_Y)^4] \sigma_Y^{-4}$$

Calculating variance two ways

Let Y be monthly earnings, measured in thousands of euros:

y	1	2	4
$\Pr(Y = y)$	.25	.50	.25

Definition

$$\mathbb{E}(Y) = .25(1) + .50(2) + .25(4) = 2.25$$
$$\begin{aligned}\mathrm{Var}(Y) &= .25(1 - 2.25)^2 + .50(2 - 2.25)^2 \\ &\quad + .25(4 - 2.25)^2 = 1.1875\end{aligned}$$

Shortcut

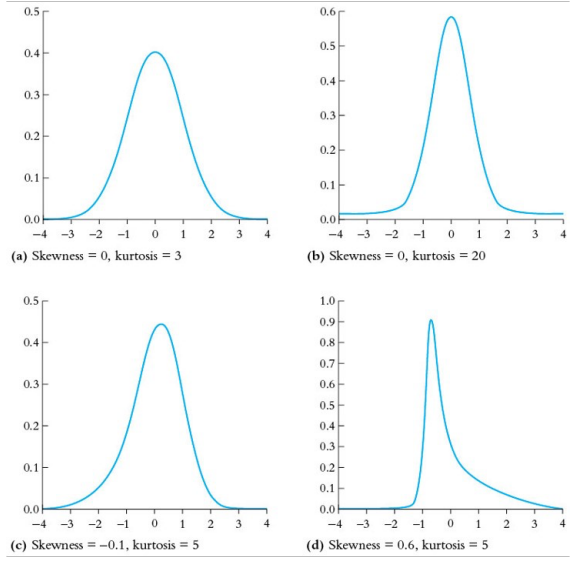
$$\mathbb{E}(Y^2) = .25(1^2) + .50(2^2) + .25(4^2) = 6.25$$
$$\begin{aligned}\mathrm{Var}(Y) &= \mathbb{E}(Y^2) - [\mathbb{E}(Y)]^2 \\ &= 6.25 - 2.25^2 = 1.1875\end{aligned}$$

$$\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab\,\mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

Examples

- $\gamma_Y = 0$: distribution is symmetric
- $\gamma_Y > 0$: skewed to the right
- $\gamma_Y < 0$: skewed to the left

- $\kappa_Y > 3$: fat tails ("leptokurtosis")



Joint Probability Distribution $\Pr(X = x, Y = y)$

Two dummy variables:

- $X \in \{0, 1\}$ — Sunny Day (N/Y)
- $Y \in \{0, 1\}$ — Ice Cream (N/Y)

Pr	X=0	X=1	Total
Y=0	.15	.25	.4
Y=1	.05	.55	.6
Total	.2	.8	1

Marginal Probability distribution:

$$\Pr(Y = y) = \sum_i \Pr(X = x_i, Y = y)$$

Conditional Probability distribution:

$$\Pr(Y = y \mid X = x) = \frac{\Pr(X = x, Y = y)}{\Pr(X = x)}$$

$$\Rightarrow \Pr(X = x, Y = y) = \Pr(Y = y \mid X = x) \Pr(X = x)$$
$$= \Pr(X = x \mid Y = y) \Pr(Y = y)$$

Working with the joint distribution

Pr	X=0	X=1	Total
Y=0	.15	.25	.4
Y=1	.05	.55	.6
Total	.2	.8	1

$$\Pr(Y=1) = .05 + .55 = .60$$
$$\Pr(X=1) = .25 + .55 = .80$$
$$\Pr(X=1, Y=1) = .55$$
$$\Pr(Y=1 \mid X=1) = \frac{\Pr(X=1, Y=1)}{\Pr(X=1)} = \frac{.55}{.80} = .6875$$
$$\Pr(X=1 \mid Y=1) = \frac{\Pr(X=1, Y=1)}{\Pr(Y=1)} = \frac{.55}{.60} \simeq .9167$$

Same cell, different denominator: conditioning changes the reference group.

Conditional and Iterated Expectations

Two dummy variables:

- $X \in \{0, 1\}$ — Sunny Day (N/Y)
- $Y \in \{0, 1\}$ — Ice Cream (N/Y)

Pr	X=0	X=1	Total
Y=0	.15	.25	.4
Y=1	.05	.55	.6
Total	.2	.8	1

Conditional Expectation:

$$\mathbb{E}(Y \mid X = x) = \sum_{i=1}^{k} y_i \Pr(Y = y_i \mid X = x)$$

Law of Iterated Expectations:

$$\begin{aligned} \mathbb{E}(Y) &= \mathbb{E}[\mathbb{E}(Y \mid X)] \\ &\equiv \sum_i \mathbb{E}(Y \mid X = x_i) \Pr(X = x_i) \end{aligned}$$

Calculating the law of iterated expectations

Because Y is binary, $\mathbb{E}(Y \mid X = x) = \Pr(Y = 1 \mid X = x)$.

$$\mathbb{E}(Y \mid X = 1) = \frac{.55}{.80} = .6875$$
$$\mathbb{E}(Y \mid X = 0) = \frac{.05}{.20} = .25$$

$$\begin{aligned}\mathbb{E}[\mathbb{E}(Y \mid X)] &= \mathbb{E}(Y \mid X = 1)\Pr(X = 1) \\ &\quad + \mathbb{E}(Y \mid X = 0)\Pr(X = 0) \\ &= .6875(.80) + .25(.20) \\ &= .55 + .05 = .60.\end{aligned}$$

This matches the marginal mean:

$$\mathbb{E}(Y) = 1\Pr(Y = 1) + 0\Pr(Y = 0) = .60.$$

Independence

- Two random variables X and Y are independent when the occurrence of one variable does not influence the value or affect the occurrence of the other.

$$\Pr(Y = y \mid X = x) = \Pr(Y = y) \quad \Longleftrightarrow \quad \Pr(X = x, Y = y) = \Pr(Y = y) \Pr(X = x)$$

- The joint probability distribution of X and Y can be expressed as the product of their individual probability distributions

- For example, think of the outcome of throwing two dice, X and Y.

An independent joint distribution

Let $X = 1$ mean die 1 shows a six, and let $Y = 1$ mean die 2 is even.

Pr	X=0	X=1	Total
Y=0	$\frac{5}{12}$	$\frac{1}{12}$	$\frac{1}{2}$
Y=1	$\frac{5}{12}$	$\frac{1}{12}$	$\frac{1}{2}$
Total	$\frac{5}{6}$	$\frac{1}{6}$	1

$$\Pr(X = 1, Y = 1) = \frac{1}{12} = \frac{1}{6}\frac{1}{2} = \Pr(X = 1)\Pr(Y = 1),$$
$$\Pr(Y = 1 \mid X = 1) = \frac{\Pr(X = 1, Y = 1)}{\Pr(X = 1)} = \frac{1/12}{1/6} = \frac{1}{2} = \Pr(Y = 1),$$

Since independence makes the joint distribution factorize,

$$\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y) \quad \text{and here} \quad \mathbb{E}(XY) = \frac{1}{12} = \mathbb{E}(X)\mathbb{E}(Y).$$

Covariance

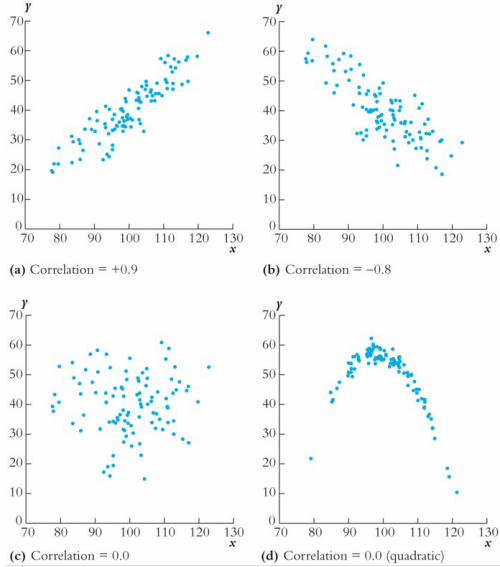
- The covariance between X and Y, σ_{XY} or $\text{Cov}(X, Y)$, is the average product of how two variables differ from their means. It measures how X and Y move together.

$$\begin{aligned}\sigma_{XY} &= \mathbb{E}[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))] \\ &= \sum_{i=1}^{k}\sum_{j=1}^{l}(x_i - \mu_X)(y_j - \mu_Y)\,\Pr(X = x_i, Y = y_j)\end{aligned}$$

- A positive covariance suggests that when one variable is above its mean, the other tends to be above its mean too. When X and Y are independent, $\sigma_{XY} = 0$.
- The converse is not true in general: zero covariance does not necessarily imply independence.
- However, interpreting covariance is tricky due to scale differences...

Correlation

- To clarify the relationship, covariance is normalized by the variables' standard deviations.
- This normalized coefficient is the correlation, $\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} \in [-1, 1]$.



Random Sampling

- In simple random sampling, n objects ($Y_1, Y_2, ..., Y_n$) are randomly drawn from a population (e.g., days, individuals, firms, etc.) such that each object has an equal and independent chance of being included in the sample
- It aims to form a representative sample to draw valid conclusions about the population's characteristics based on observed sample data.
- Because $Y_1, Y_2, ..., Y_n$ are randomly drawn from the same population, the marginal distribution of Y_i is the same for each i: $Y_1, Y_2, ..., Y_n$ are identically distributed
- Because $Y_1, Y_2, ..., Y_n$ are randomly drawn, knowing the value of Y_i does not provide information about $Y_{j \neq i}$: $Y_1, Y_2, ..., Y_n$ are independently distributed
- When $Y_1, Y_2, ..., Y_n$ are randomly drawn from the same distribution, they are independently and identically distributed (i.i.d.). Then, the expected value of every draw is the population mean and the variance of every draw is the population variance.

Sample Distribution Mean and Dispersion

- Because the sample is drawn at random, the sample average (and other statistics), $\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$, is also random.
- Because $\overline{Y}$ is random, it has a probability distribution. The distribution of $\overline{Y}$ is called the sampling distribution of $\overline{Y}$, i.e., it is the probability distribution associated with possible values of $\overline{Y}$ that could be computed for different random samples of size n.
- The expected value of the sampling distribution of $\overline{Y}$ is the population mean: $\mathbb{E}(\overline{Y}) = \mu_Y$
- The variance of the sampling distribution of $\overline{Y}$ is:

$$\sigma^2_{\overline{Y}} = \text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}(Y_i) + \underbrace{\frac{1}{n^2}\sum_{i=1}^{n}\sum_{\substack{j=1 \\ j\neq i}}^{n}\text{Cov}(Y_i, Y_j)}_{=0 \text{ (because i.i.d.)}}$$

$$= \frac{1}{n^2} \times n\sigma^2_Y + 0 = \frac{1}{n}\sigma^2_Y.$$

- The standard deviation of the sampling distribution is therefore given by $\frac{\sigma_Y}{\sqrt{n}}$

Asymptotic properties ($n \to \infty$) of the sampling distribution (1/2)

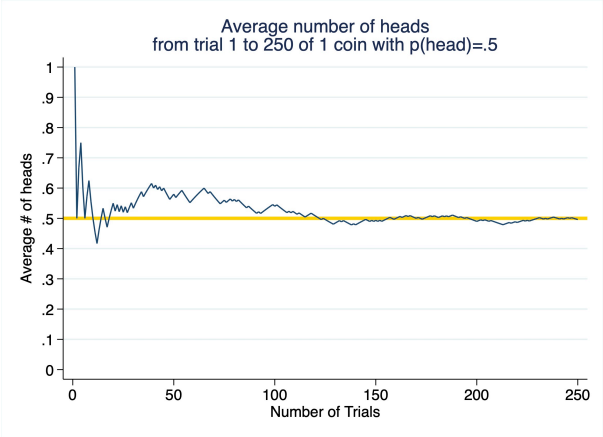
- If $Y_1, Y_2, ..., Y_n$ are drawn randomly from the same population (i.i.d.) and large outliers are unlikely, then $\overline{Y}$ will approximate $\mathbb{E}(Y)$ as n grows
  - In our previous example, the expected value for a sunny day is .8. If we were to randomly draw days from the entire history, the average number of sunny days converges to .8 as we draw more days from history.
  - If you flip it a lot of times, the average number of heads you get gets closer to 50%, which is what we expect from a fair coin.
- In this case, we say that $\overline{Y}$ converges in probability to $\mathbb{E}(Y)$
  - The probability that $\overline{Y}$ equals $\mathbb{E}(Y)$ converges to 1 as n grows (as $\lim_{n\to\infty} \sigma^2_{\overline{Y}} = 0$)

(Weak) Law of Large Numbers

If Y_i are independent and identically distributed and $| \mathbb{E}(Y) | < \infty$, then as $n \to \infty$,

$$\overline{Y} = \frac{1}{n} \sum_{i=1}^{n} Y_i \xrightarrow{p} \mathbb{E}(Y) \equiv \mu_Y$$

Example – tossing a coin



Asymptotic properties ($n \to \infty$) of the sampling distribution (2/2)

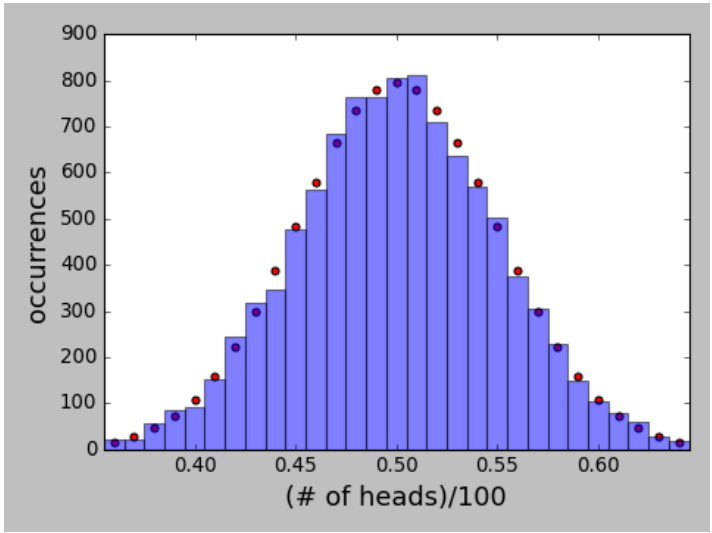
- Random sample averages behave in a predictable pattern.
- If $Y_1, Y_2, ..., Y_n$ are drawn randomly from the same population (i.i.d.) and large outliers are unlikely, then $\overline{Y}$ is approximately distributed as $\mathcal{N}(\mu_Y, \sigma_Y^2/n)$ as n grows
  - When you randomly gather a large group of people from UPF and calculate their average height, repeating this with many groups, the resulting sample averages distribution will form a bell-shaped curve, regardless of the actual height distribution at UPF.
- $\overline{Y} \sim \mathcal{N}(\mu_Y, \sigma_{\overline{Y}}^2)$ exactly when $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, and approximately when Y follows another distribution
- Very powerful because it allows us to make better predictions and draw conclusions about populations based on smaller samples

Central Limit Theorem

If Y_i are independent and identically distributed and $0 < \sigma_Y^2 < \infty$, then as $n \to \infty$,

$$\frac{\overline{Y} - \mu_Y}{\sigma_Y/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1)$$

Example – tossing a coin ($n = 100$ flips, 10K times)



Estimators

- Imagine that μ_Y is unknown and we want to estimate it. The sample average is a natural choice:

$$\overline{Y} = \frac{Y_1 + \cdots + Y_n}{n}.$$

  - This formula is an *estimator*: a rule applied to sample data.
- Before observing the sample, the Y_i's are random, so $\overline{Y}$ is random: different samples can give different values.
- After observing one sample, the estimator gives an *estimate*. For example,

$$(y_1, y_2, y_3) = (1.70, 1.80, 1.75) \quad \Longrightarrow \quad \overline{y} = 1.75.$$

  - The estimator is the rule; the estimate is the resulting number.
- How to determine if an estimator is better than another? Let $\hat{\mu}_Y$ and $\tilde{\mu}_Y$ be two estimators of μ_Y.
  1. $\hat{\mu}_Y$ is *unbiased* if $\mathbb{E}(\hat{\mu}_Y) = \mu_Y$
  2. $\hat{\mu}_Y$ is *consistent* if $\hat{\mu}_Y \xrightarrow{p} \mu_Y$
  3. $\hat{\mu}_Y$ is *more efficient* than $\tilde{\mu}_Y$ if $\text{Var}(\hat{\mu}_Y) \leq \text{Var}(\tilde{\mu}_Y)$

Why is the sample mean efficient?

- *Efficiency* means precision: among unbiased estimators, prefer the one with the smallest variance.
- **Step 1: Cramér–Rao gives a lower bound.** For any unbiased estimator $\hat{\mu}_Y$,

$$\text{Var}(\hat{\mu}_Y) \geq \frac{1}{\mathcal{I}_n(\mu_Y)}.$$

- For n independent observations from a normal population with variance σ_Y^2,

$$\mathcal{I}_n(\mu_Y) = \frac{n}{\sigma_Y^2} \quad \Longrightarrow \quad \text{Var}(\hat{\mu}_Y) \geq \frac{\sigma_Y^2}{n}.$$

- **Step 2: the sample mean reaches the bound:**

$$\mathbb{E}(\overline{Y}) = \mu_Y, \qquad \text{Var}(\overline{Y}) = \frac{\sigma_Y^2}{n}.$$

- **Conclusion:** no unbiased estimator can have a smaller variance. Therefore $\overline{Y}$ is efficient.

BLUE and least squares

1. **BLUE: comparing estimators**
   - $\overline{Y}$ is the **B**est **L**inear **U**nbiased **E**stimator of μ_Y.
   - **Linear:** a weighted sum of $Y_1, \ldots, Y_n$.
   - **Unbiased:** its expected value is μ_Y.
   - **Best:** smallest variance among such estimators.

2. **Least squares: minimizing errors**
   - Choose $\hat{\mu}_Y$ to minimize the sum of squared deviations:

$$\sum_{i=1}^{n}(Y_i - \hat{\mu}_Y)^2.$$

   - First-order condition:

$$-2\sum_{i=1}^{n}(Y_i - \hat{\mu}_Y) = 0 \quad \Longrightarrow \quad \hat{\mu}_Y = \overline{Y}.$$

Under the standard i.i.d. assumptions, both approaches give the sample mean.

Material I

– **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapters 1 - 2